

Saturday Test — No. 5

Take your time. Show your work where it helps.

Name: _____

Date: _____

Time: 15 min

Numbers and Shapes

1. Using each of the digits **5, 0, 8, 3** exactly once, write the **smallest** four-digit number you can.
2. List every **perfect square** between 1 and 30. (A perfect square is a number you get by multiplying a whole number by itself.)

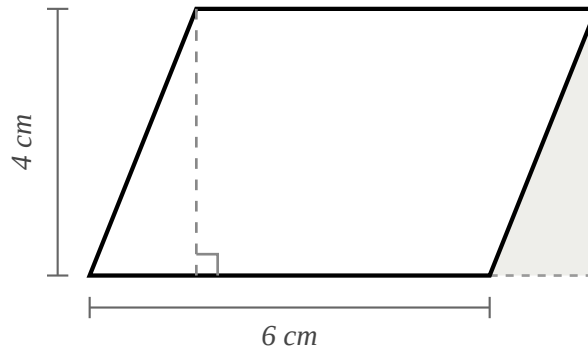
3. Take 612.

- (a) Is it divisible by 3?
- (b) Now add up its digits. Is *that sum* divisible by 3?

4. Take 514.

- (a) Is it divisible by 3?
- (b) Now add up its digits. Is *that sum* divisible by 3?

5. Find the area of this parallelogram.



Counting Carefully

6. You have only 3¢ stamps and 5¢ stamps (as many of each as you want). Which amounts from 1¢ to 15¢ can you **not** make? List them all.

7. A drawer holds **10 red socks** and **10 blue socks**, all mixed up. You reach in without looking and pull socks out one at a time. What is the **smallest** number of socks you must pull out to be *sure* you have **two red socks**?

Reasoning

8. **10!** means $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10$.
How many **zeros** are at the end of this number?

Answer Key — for parents

Numbers and Shapes

1. The smallest four-digit number must start with the smallest non-zero digit available (the leading digit can't be 0). That's **3**. After placing 3, arrange the remaining digits 0, 5, 8 in ascending order: 058. Answer: **3058**.

2. $1 \times 1 = 1$, $2 \times 2 = 4$, $3 \times 3 = 9$, $4 \times 4 = 16$, $5 \times 5 = 25$. Next would be $6 \times 6 = 36$, which is past 30. Answer: **1, 4, 9, 16, 25**.

3. (a) $612 \div 3 = 204$ with no remainder, so **yes**, 612 is divisible by 3. (b) $6 + 1 + 2 = 9$, and $9 \div 3 = 3$ with no remainder, so **yes**, the digit-sum is also divisible by 3.

4. (a) $514 \div 3 = 171$ remainder 1, so **no**, 514 is not divisible by 3. (b) $5 + 1 + 4 = 10$, and $10 \div 3 = 3$ remainder 1, so **no**, the digit-sum is also not divisible by 3.

Pattern to point out: the two methods agree on both numbers, and they always do. A whole number is divisible by 3 exactly when the sum of its digits is divisible by 3 — a real shortcut that works for any number, no matter how large.

5. If you cut along the dashed line and slide the triangle from the left side over to the slot on the right, the parallelogram becomes a 6×4 rectangle — same area. So the area is $\text{base} \times \text{height} = 6 \times 4 = \mathbf{24 \text{ cm}^2}$. The lesson: for any parallelogram, $\text{area} = \text{base} \times \text{perpendicular height}$ (the slant doesn't matter).

Counting Carefully

6. Reachable amounts: 3, 5, 6 ($=3+3$), 8 ($=3+5$), 9 ($=3+3+3$), 10 ($=5+5$), 11 ($=3+3+5$), 12 ($=3+3+3+3$), 13 ($=3+5+5$), 14 ($=3+3+3+5$), 15 ($=5+5+5$ or 3×5). Cannot be made: **1¢, 2¢, 4¢, 7¢**. *Worth noticing:* once you reach 8¢, you can also get 9¢ and 10¢, and after that you can keep adding 3¢ forever — so every amount from 8¢ upward is reachable. 7¢ is the last impossible amount.

7. Worst case: you pull out every blue sock first (all 10), then the next two pulls must be red. So you need **12** socks to be sure. (With 11 socks you might have all 10 blues plus 1 red — still only one red.)

Reasoning

8. A zero at the end of a number comes from a factor of 10, and $10 = 2 \times 5$. So the number of trailing zeros equals the number of (2,5) pairs you can pull out of the product. Factors of 5 in $1 \times 2 \times \dots \times 10$: only **5** and **10** contribute (one 5 each). Factors of 2: plenty — 2, 4, 6, 8, 10 give more than enough 2s to pair with. So exactly **2** trailing zeros. (Check: $10! = 3,628,800$.) *The big idea:* you didn't need to compute $10!$ at all — the 5s are the bottleneck.